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Database Systems
Chapter 7 – Case Study 2 (Projects Schema)

```
=====
Exercise 7.24: Create the Projects Schema
=====
```

```
mysql> use ProjectDB;
Database changed
mysql> show tables;
+-----+
| Tables_in_projectdb |
+-----+
| department           |
| employee              |
+-----+
2 rows in set (0.02 sec)

mysql> drop table employee;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> show tables;
+-----+
| Tables_in_projectdb |
+-----+
| department           |
+-----+
1 row in set (0.00 sec)
```

```
mysql> create table employee (
  -> empno int primary key,
  -> fname varchar(30) not null,
  -> lname varchar(30) not null,
  -> address varchar(100),
  -> dob date,
  -> sex char(1)
  -> check (sex in ('M', 'F')),
  -> position varchar(30)
  -> check (position in (
  -> 'Manager',
  -> 'Team Leader',
  -> 'Analyst',
  -> 'Software Developer'
  -> )),
  -> deptno int,
```

```
-> constraint fk_employee_department
-> foreign key (deptno)
-> references department(deptno)
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> alter table department
-> add constraint FK_Department_Manager
-> foreign key (mgrEmpNo)
-> references employee(empno);
```

Query OK, 0 rows affected (0.19 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> create table Project (
-> projectNo int primary key,
-> projectName varchar(50) not null,
-> deptNo int not null,
-> constraint FK_Project_Department
-> foreign key(deptNo)
-> references Department(deptNo)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> create table workOn(
-> empno int,
-> projNo int,
-> dateWorked date,
-> hourWorked int
-> check(hoursWorked between 0 and 40),
-> primary key(empno, projNo, dateWorked),
-> constraint FK_WorksOn_Employee
-> foreign key(empno)
-> references Employee(empno),
-> constraint FK_WorksOn_Project
-> foreign key(projNo)
-> references Project(projNo)
-> );
```

ERROR 3813 (HY000): Column check constraint 'workon_chk_1' references other column.

```
mysql> create table workOn (
-> empno int,
-> projNo int,
-> dateWorked date,
-> hourWorked int,
-> check (hourWorked between 0 and 40),
-> primary key (empno, projNo, dateWorked),
-> constraint fk_workson_employee
-> foreign key (empno)
-> references employee(empno),
-> constraint fk_workson_project
-> foreign key (projNo)
```

```
-> references project(projNo)
-> );
```

ERROR 3734 (HY000): Failed to add the foreign key constraint. Missing column 'projNo' for constraint 'fk_workson_project' in the referenced table 'project'

```
mysql> describe Project;
```

Field	Type	Null	Key	Default	Extra
projectNo	int	NO	PRI	NULL	
projectName	varchar(50)	NO		NULL	
deptNo	int	NO	MUL	NULL	

3 rows in set (0.01 sec)

```
mysql> ^C
```

```
mysql> create table workOn (
```

```
-> empno int,
-> projectNo int,
-> dateWorked date,
-> hourWorked int,
-> check (hourWorked between 0 and 40),
-> primary key (empno, projectNo, dateWorked),
-> constraint fk_workson_employee
-> foreign key (empno)
-> references employee(empno),
-> constraint fk_workson_project
-> foreign key (projectNo)
-> references project(projectNo)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> describe department;
```

Field	Type	Null	Key	Default	Extra
deptNo	int	NO	PRI	NULL	
deptName	varchar(50)	NO		NULL	
mgrEmpNo	int	YES	MUL	NULL	

3 rows in set (0.00 sec)

=====

Exercise 7.25 CREATE VIEW female_manager_projects

=====

```
mysql> create view female_manager_projects as
```

```
-> select
->     p.projectNo,
->     p.projectName,
->     d.deptNo,
```

```

->     d.deptName,
->     e.empNo as managerID,
->     e.fName,
->     e.lName
-> from project p
-> join department d
->     on p.deptNo = d.deptNo
-> join employee e
->     on d.mgrEmpNo = e.empNo
-> where e.sex = 'F'
-> order by p.projectNo;
Query OK, 0 rows affected (0.02 sec)

```

```

mysql> select * from Female_Manager_Projects;
Empty set (0.00 sec)

```

```

mysql> select * from Department;
Empty set (0.00 sec)

```

```

mysql> select * from employee;
Empty set (0.00 sec)

```

```

mysql> select * from Project;
Empty set (0.00 sec)

```

```

mysql> INSERT INTO Department (deptNo, deptName)
-> VALUES
-> (1, 'Information Technology'),
-> (2, 'Finance'),
-> (3, 'Human Resources'),
-> (4, 'Marketing'),
-> (5, 'Operations');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

mysql> INSERT INTO Employee
-> (empNo, fName, lName, address, DOB, sex, position, deptNo)
-> VALUES
-> (101, 'Mary', 'Mensah', 'Accra', '1990-03-15', 'F', 'Manager', 1),
-> (102, 'John', 'Doe', 'Kumasi', '1992-07-20', 'M', 'Analyst', 1),
-> (103, 'Linda', 'Boateng', 'Takoradi', '1989-05-18', 'F', 'Manager', 2),
-> (104, 'Peter', 'Owusu', 'Cape Coast', '1993-08-12', 'M', 'Team Leader', 2),
-> (105, 'Grace', 'Asante', 'Sunyani', '1991-01-30', 'F', 'Manager', 3),
-> (106, 'Michael', 'Amoah', 'Tamale', '1994-04-22', 'M', 'Software Developer',
3),
-> (107, 'Sarah', 'Antwi', 'Ho', '1990-12-01', 'F', 'Manager', 4),
-> (108, 'David', 'Mensah', 'Koforidua', '1995-07-10', 'M', 'Analyst', 4),
-> (109, 'Esther', 'Nyarko', 'Wa', '1988-09-14', 'F', 'Manager', 5),
-> (110, 'Daniel', 'Boadu', 'Bolgatanga', '1996-02-05', 'M', 'Software
Developer', 5);

```

Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> UPDATE Department SET mgrEmpNo = 101 WHERE deptNo = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Department SET mgrEmpNo = 103 WHERE deptNo = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Department SET mgrEmpNo = 105 WHERE deptNo = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Department SET mgrEmpNo = 107 WHERE deptNo = 4;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Department SET mgrEmpNo = 109 WHERE deptNo = 5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> INSERT INTO Project (projNo, projName, deptNo)
-> VALUES
-> (1001, 'Student Portal', 1),
-> (1002, 'Payroll System', 2),
-> (1003, 'Recruitment System', 3),
-> (1004, 'Digital Marketing Platform', 4),
-> (1005, 'Inventory Management System', 5);
ERROR 1054 (42S22): Unknown column 'projNo' in 'field list'
mysql> describe project;

Field	Type	Null	Key	Default	Extra
projectNo	int	NO	PRI	NULL	
projectName	varchar(50)	NO		NULL	
deptNo	int	NO	MUL	NULL	

3 rows in set (0.00 sec)

mysql> INSERT INTO Project (projectNo, projectName, deptNo)
-> VALUES
-> (1001, 'Student Portal', 1),
-> (1002, 'Payroll System', 2),
-> (1003, 'Recruitment System', 3),
-> (1004, 'Digital Marketing Platform', 4),
-> (1005, 'Inventory Management System', 5);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

```
mysql> drop view if exists Female_Manager_Projects;
Query OK, 0 rows affected (0.01 sec)
```

```
=====
Exercise 7.26
=====
```

```
mysql> CREATE VIEW Female_Manager_Projects AS
-> SELECT
->     P.projectNo,
->     P.projectName,
->     D.deptNo,
->     D.deptName,
->     E.empNo AS ManagerID,
->     E.fName,
->     E.lName
-> FROM Project P
-> JOIN Department D
->     ON P.deptNo = D.deptNo
-> JOIN Employee E
->     ON D.mgrEmpNo = E.empNo
-> WHERE E.sex = 'F'
-> ORDER BY P.projectNo;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SHOW CREATE TABLE Employee;
```

```
+-----+-----+-----+-----+-----+-----+
| Table | Create Table
```

```
+-----+-----+-----+-----+-----+-----+
|
```

```

-----+
| Employee | CREATE TABLE `employee` (
  `empno` int NOT NULL,
  `fname` varchar(30) NOT NULL,
  `lname` varchar(30) NOT NULL,
  `address` varchar(100) DEFAULT NULL,
  `dob` date DEFAULT NULL,
  `sex` char(1) DEFAULT NULL,
  `position` varchar(30) DEFAULT NULL,
  `deptno` int DEFAULT NULL,
  PRIMARY KEY (`empno`),
  KEY `fk_employee_department` (`deptno`),
  CONSTRAINT `fk_employee_department` FOREIGN KEY (`deptno`) REFERENCES `department`
(`deptNo`),
  CONSTRAINT `employee_chk_1` CHECK ((`sex` in (_utf8mb4'M',_utf8mb4'F'))),
  CONSTRAINT `employee_chk_2` CHECK ((`position` in (_utf8mb4'Manager',_utf8mb4'Team
Leader',_utf8mb4'Analyst',_utf8mb4'Software Developer'))))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

```

```

-----+
1 row in set (0.00 sec)

```

```

mysql> SHOW CREATE TABLE Department;

```

```

+-----+
| Table      | Create Table

```

```

-----+
| Department | CREATE TABLE `department` (
  `deptNo` int NOT NULL,

```

```

`deptName` varchar(50) NOT NULL,
`mgrEmpNo` int DEFAULT NULL,
PRIMARY KEY (`deptNo`),
KEY `FK_Department_Manager` (`mgrEmpNo`),
CONSTRAINT `FK_Department_Manager` FOREIGN KEY (`mgrEmpNo`) REFERENCES `employee`
(`empno`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
-----+
-----+
-----+
-----+
-----+
1 row in set (0.00 sec)

```

```

mysql> SHOW CREATE TABLE WorksOn;
ERROR 1146 (42S02): Table 'projectdb.workson' doesn't exist
mysql> show tables;

```

```

+-----+
| Tables_in_projectdb |
+-----+
| department          |
| employee             |
| female_manager_projects |
| project              |
| workon               |
+-----+
5 rows in set (0.00 sec)

```

```

mysql> describe project;

```

```

+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| projectNo  | int       | NO   | PRI | NULL    |       |
| projectName | varchar(50) | NO   |     | NULL    |       |
| deptNo     | int       | NO   | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

```

mysql> describe workOn;

```

```

+-----+-----+-----+-----+-----+-----+
| Field      | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| empno      | int  | NO   | PRI | NULL    |       |
| projectNo  | int  | NO   | PRI | NULL    |       |
| dateWorked | date | NO   | PRI | NULL    |       |
| hourWorked | int  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

```

```

mysql> select * from project;

```


projectNo	projectName	deptNo
1001	Student Portal	1
1002	Payroll System	2
1003	Recruitment System	3
1004	Digital Marketing Platform	4
1005	Inventory Management System	5

5 rows in set (0.00 sec)

```
mysql> select * from workon;
Empty set (0.00 sec)
```

```
mysql> INSERT INTO workOn
-> (empNo, projectNo, dateWorked, hourWorked)
-> VALUES
-> (101,1001,'2026-06-20',8),
-> (102,1001,'2026-06-20',6),
-> (103,1002,'2026-06-20',7),
-> (104,1002,'2026-06-20',5),
-> (105,1003,'2026-06-20',8),
-> (106,1003,'2026-06-20',6),
-> (107,1004,'2026-06-20',8),
-> (108,1004,'2026-06-20',7),
-> (109,1005,'2026-06-20',8),
-> (110,1005,'2026-06-20',6);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> drop view if exists female_manager_projects;
Query OK, 0 rows affected (0.01 sec)
```

Exercise 7.27

```
mysql> CREATE VIEW female_manager_projects AS
-> SELECT
->     P.projectNo,
->     P.projectName,
->     D.deptNo,
->     D.deptName,
->     E.empNo AS ManagerID,
->     E.fName,
->     E.lName
-> FROM project P
-> JOIN department D
->     ON P.deptNo = D.deptNo
-> JOIN employee E
```

```

->      ON D.mgrEmpNo = E.empNo
-> WHERE E.sex = 'F'
-> ORDER BY P.projectNo;

```

Query OK, 0 rows affected (0.01 sec)

```

mysql> select * from female_manager_project;
ERROR 1146 (42S02): Table 'projectdb.female_manager_project' doesn't exist
mysql> select * from female_manager_projects;

```

```

+-----+-----+-----+-----+-----+
| projectNo | projectName | deptNo | deptName |
+-----+-----+-----+-----+
| 1001 | Student Portal | 1 | Information Technology |
101 | Mary | Mensah |
| 1002 | Payroll System | 2 | Finance |
103 | Linda | Boateng |
| 1003 | Recruitment System | 3 | Human Resources |
105 | Grace | Asante |
| 1004 | Digital Marketing Platform | 4 | Marketing |
107 | Sarah | Antwi |
| 1005 | Inventory Management System | 5 | Operations |
109 | Esther | Nyarko |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

```
mysql> show tables;
```

```

+-----+
| Tables_in_projectdb |
+-----+
| department |
| employee |
| female_manager_projects |
| project |
| workon |
+-----+
5 rows in set (0.00 sec)

```

```
mysql> describe employee;
```

```

+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| empno | int | NO | PRI | NULL | |
| fname | varchar(30) | NO | | NULL | |
| lname | varchar(30) | NO | | NULL | |
| address | varchar(100) | YES | | NULL | |
| dob | date | YES | | NULL | |
| sex | char(1) | YES | | NULL | |

```

position	varchar(30)	YES		NULL	
deptno	int	YES	MUL	NULL	

8 rows in set (0.00 sec)

mysql> describe Department;

Field	Type	Null	Key	Default	Extra
deptNo	int	NO	PRI	NULL	
deptName	varchar(50)	NO		NULL	
mgrEmpNo	int	YES	MUL	NULL	

3 rows in set (0.00 sec)

mysql> select * from department;

deptNo	deptName	mgrEmpNo
1	Information Technology	101
2	Finance	103
3	Human Resources	105
4	Marketing	107
5	Operations	109

5 rows in set (0.00 sec)

mysql> select * from employee;

empno	fname	lname	address	dob	sex	position	deptno
101	Mary	Mensah	Accra	1990-03-15	F	Manager	1
102	John	Doe	Kumasi	1992-07-20	M	Analyst	1
103	Linda	Boateng	Takoradi	1989-05-18	F	Manager	2
104	Peter	Owusu	Cape Coast	1993-08-12	M	Team Leader	2
105	Grace	Asante	Sunyani	1991-01-30	F	Manager	3
106	Michael	Amoah	Tamale	1994-04-22	M	Software Developer	3
107	Sarah	Antwi	Ho	1990-12-01	F	Manager	4
108	David	Mensah	Koforidua	1995-07-10	M	Analyst	4

109	Esther	Nyarko	Wa	1988-09-14	F	Manager
5						
110	Daniel	Boadu	Bolgatanga	1996-02-05	M	Software Developer
5						

```

+-----+-----+-----+-----+-----+-----+-----+
-----+

```

10 rows in set (0.00 sec)

```
mysql> select * from project;
```

projectNo	projectName	deptNo
1001	Student Portal	1
1002	Payroll System	2
1003	Recruitment System	3
1004	Digital Marketing Platform	4
1005	Inventory Management System	5

5 rows in set (0.00 sec)

```
mysql> select * from workon;
```

empno	projectNo	dateWorked	hourWorked
101	1001	2026-06-20	8
102	1001	2026-06-20	6
103	1002	2026-06-20	7
104	1002	2026-06-20	5
105	1003	2026-06-20	8
106	1003	2026-06-20	6
107	1004	2026-06-20	8
108	1004	2026-06-20	7
109	1005	2026-06-20	8
110	1005	2026-06-20	6

10 rows in set (0.00 sec)

```
mysql> create view employee_Project_Work as select
```

```

-> E.empno,
-> F.fName,\c

```

```
mysql> create view employee_Project_Work as select
```

```

-> E.empNo,
-> F.fName,
-> E.lName,
-> P.projectName,
-> W.hourWorked
-> from Employee E join WorkOn W
-> on E.empNo = W.empNo
-> join Project P
-> on W.projectNo = P.projectNo;

```

ERROR 1054 (42S22): Unknown column 'F.fName' in 'field list'

mysql> create view employee_project_work as

```
-> select
->     E.empNo,
->     E.fName,
->     E.lName,
->     P.projectName,
->     W.hourWorked
-> from employee E
-> join workOn W
->     on E.empNo = W.empNo
-> join project P
->     on W.projectNo = P.projectNo;
```

Query OK, 0 rows affected (0.01 sec)

mysql> select * from employee_project_work;

empNo	fName	lName	projectName	hourWorked
101	Mary	Mensah	Student Portal	8
102	John	Doe	Student Portal	6
103	Linda	Boateng	Payroll System	7
104	Peter	Owusu	Payroll System	5
105	Grace	Asante	Recruitment System	8
106	Michael	Amoah	Recruitment System	6
107	Sarah	Antwi	Digital Marketing Platform	8
108	David	Mensah	Digital Marketing Platform	7
109	Esther	Nyarko	Inventory Management System	8
110	Daniel	Boadu	Inventory Management System	6

10 rows in set (0.00 sec)

mysql> create view EmpProject(empNo, projNo, totalHours)

```
-> as select w.empNo, w.projNo, sum(hoursWorked)
-> ...
-> \c
```

mysql> create view emproject (empNo, projNo, totalHours) as

```
-> select w.empNo, w.projNo, sum(w.hourWorked) from workon w group by w.empNo,
w,projNo;
```

ERROR 1054 (42S22): Unknown column 'w.projNo' in 'field list'

mysql> create view emproject (empNo, projectNo, totalHours) as

```
-> select
->     w.empNo,
->     w.projectNo,
->     sum(w.hourWorked)
-> from workOn w
-> group by
->     w.empNo,
->     w.projectNo;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> show full tables where table_type = 'view';
Empty set (0.00 sec)
```

```
mysql> show create view emproject;
```

Handwriting practice lines with a sample sentence. The lines are dashed and solid. The sample sentence is: "View Create View".

collation	connection		character_set_client	
-----------	------------	--	----------------------	--

```
+-----+
| empproject | CREATE ALGORITHM=UNDEFINED DEFINER=`root`@`localhost` SQL SECURITY
DEFINER VIEW `empproject` (`empNo`,`projectNo`,`totalHours`) AS select `w`.`empno` AS
`empNo`,`w`.`projectNo` AS `projectNo`,sum(`w`.`hourWorked`) AS `sum(w.hourWorked)`
from `workon` `w` group by `w`.`empno`,`w`.`projectNo` | utf8mb4
utf8mb4_0900_ai_ci |
+-----+
1 row in set (0.00 sec)
```

```
mysql> show full tables;
```

Tables_in_projectdb	Table_type
department	BASE TABLE
employee	BASE TABLE
employee_project_work	VIEW
emproject	VIEW
female_manager_projects	VIEW
project	BASE TABLE
workon	BASE TABLE

7 rows in set (0.00 sec)

```
mysql> CREATE VIEW Employee_Project_Work AS
-> SELECT
->     E.empNo,
->     E.fName,
```

```

->     E.lName,
->     P.projectName,
->     W.hourWorked
-> FROM Employee E
-> JOIN WorkOn W
->     ON E.empNo = W.empNo
-> JOIN Project P
->     ON W.projectNo = P.projectNo;
ERROR 1050 (42S01): Table 'Employee_Project_Work' already exists
mysql> SHOW FULL TABLES
-> WHERE Tables_in_projectdb = 'employee_project_work';

```

```

+-----+
| Tables_in_projectdb | Table_type |
+-----+
| employee_project_work | VIEW      |
+-----+
1 row in set (0.00 sec)

```

```

mysql> CREATE VIEW EmpProject AS
-> SELECT
->     W.empNo,
->     W.projectNo,
->     SUM(W.hourWorked) AS totalHours
-> FROM Employee E
-> JOIN WorkOn W
->     ON E.empNo = W.empNo
-> JOIN Project P
->     ON P.projectNo = W.projectNo
-> GROUP BY W.empNo, W.projectNo;
Query OK, 0 rows affected (0.01 sec)

```

```

mysql> select * from EmpProject;
+-----+
| empNo | projectNo | totalHours |
+-----+
| 101   | 1001      | 8          |
| 102   | 1001      | 6          |
| 103   | 1002      | 7          |
| 104   | 1002      | 5          |
| 105   | 1003      | 8          |
| 106   | 1003      | 6          |
| 107   | 1004      | 8          |
| 108   | 1004      | 7          |
| 109   | 1005      | 8          |
| 110   | 1005      | 6          |
+-----+
10 rows in set (0.00 sec)

```

```

mysql> select * from projectNo
-> fr\c

```

```
mysql> select projectNo
-> from EmProject where projectNo = 1001;
```

```
+-----+
| projectNo |
+-----+
|      1001 |
|      1001 |
+-----+
```

2 rows in set (0.00 sec)

```
mysql> select count(projectNo)
-> from EmProject
-> where empNo = 101;
```

```
+-----+
| count(projectNo) |
+-----+
|                1 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select empNo, totalHours
-> from EmProject
-> group by empNo;
```

ERROR 1055 (42000): Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'projectdb.EmProject.totalHours' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by

```
mysql> select empNo,
-> sum(totalHours) as totalHours
-> from EmProject
-> group by empNo;
```

```
+-----+-----+
| empNo | totalHours |
+-----+-----+
| 101   | 8          |
| 102   | 6          |
| 103   | 7          |
| 104   | 5          |
| 105   | 8          |
| 106   | 6          |
| 107   | 8          |
| 108   | 7          |
| 109   | 8          |
| 110   | 6          |
+-----+-----+
```

10 rows in set (0.00 sec)

```
mysql> show databases;
```

```
+-----+
| Database |
+-----+
```



```

+-----+
| companydb      |
| contact_db     |
| contacts_db    |
| customer_db    |
| elitevideo     |
| information_schema |
| ministryfinances |
| mysql          |
| nilehub        |
| niletech       |
| niletech_hub   |
| performance_schema |
| projectdb      |
| ra_practice    |
| sys            |
+-----+
15 rows in set (0.01 sec)

```

```

mysql> select database();
+-----+
| database() |
+-----+
| projectdb  |
+-----+
1 row in set (0.00 sec)

```

```
mysql>
```

```

=====
Exercise 7.28
=====

```

The materialized view ExpensiveParts stores the part numbers of all parts with a cost greater than £1000. Since the data is stored physically, the view must be updated whenever changes are made to the Part table to ensure it remains accurate.

If a new part is inserted and its cost is more than £1000, its part number should be added to the materialized view. If an existing part's cost is updated from £1000 or less to above £1000, the part number should also be added. If the cost is reduced to £1000 or below, the part number should be removed from the view, but only if there are no other contracts for the same part with a cost greater than £1000. Likewise, if a record is deleted, the system should check whether another record for the same part still satisfies the condition before removing it from the view.

The view can be maintained without accessing the Part table in simple cases, such as when a new expensive part is inserted because the system already knows that the new record qualifies for the view. However, when a part is updated or deleted, the Part table may need to be checked to determine whether another contract for the same part still has a cost greater than £1000. This ensures that only eligible part numbers

remain in the materialized view.